

Model Settings and Output Files

Documentation Overview

November 18, 2025

Model version: 1.0.0

This document provides an overview of the key configuration parameters and output files used in the energy system optimization model FEM.

1 Model Settings

All settings in the model have a sensible default value. Any of these defaults can be overridden by the user by adding the corresponding setting item to the scenario setup CSV which is the file referenced by the *target_csv* variable in *scenarios.py*.

To help users, we distinguish between two groups: primary settings and secondary settings. Primary settings are the parameters commonly modified through the standard settings file; they are intended to be easy to change and typically do not require deep knowledge of the model. Secondary settings are more advanced parameters that also have defaults but are usually only added to the settings file when needed — overriding them may require additional code changes, extra input files, or further testing.

1.1 Primary settings

The settings file to be used is defined in *scenarios.py* at the definition of the variable *target_csv*. If those settings are removed from the file, the default value is used.

Setting	Description
<i>Item</i>	Scenario's and resulting output folder's name.
<i>sub_secn</i>	Subscenario's name. If a scenario is supposed to have several subscenarios, <i>Item</i> must be the same but <i>sub_secn</i> must be different.
<i>weight_in_objective_fcn</i>	The weight of each subscenario. It must add up to 1 across all subscenarios for each scenario.
<i>NTC_CH_ratio</i>	Scaling factor for the Swiss NTC. Default: 1.
<i>limit_fuel_import_CH</i>	Scaling factor for the upper limit of the Swiss fuel import according to the contents of the file <i>fuel_limits.csv</i> . If no limit is desired, it can be set to "False". Default: False.
<i>weather_year</i>	Weather year used for demand and renewable generation profiles. Available years are 1995, 2008 and 2009. Default: 1995.

Setting	Description
run_year	Target year of the model run in terms of the renewable expansion, etc. Available years are 2035 and 2050. Default: 2050.
t_start	Starting hour of the simulation. Hour 6553 corresponds to the beginning of October. Starting here aligns with modeling a hydro year. Default: 6553.
t_end	Last hour of the simulation. If $t_{start} > t_{end}$, the simulation will continue with t_1 after t_8760. Default: 6552.
electrolyzer_demand_reduction_coefficient	Scaling factor that modifies the annual electricity demand of all electrolyzers but the Swiss ones. These demands can be found in <i>electrolyzer_net_demand.csv</i> . Default: 1.
flexible_household_heatpump_share	Share of households that operate their heat pumps in a flexible way. Default: 0.5.
V2G_share_of_flexibly_charging_EV	Share of EVs that do bidirectional charging. Default: 0.5.
reduce_inflex_demand_by_[MWh]	Inflexible demand reduction applied to CH. A negative value can increase the demand. Default: 0.
reduce_DH_demand_by_[MWh]	District heating demand reduction applied. A negative value can increase the demand. Default: 0.
ror_annual_TWh	The annual ror generation is scaled to this value. If "False", the raw data is not scaled. Default: False.
hydro_inflow_TWh	The annual open psp generation is scaled to this value. If "False", the raw data is not scaled. Default: False.
pump_capacity_GW	Controls how much power the large Swiss psp should have. If "False", the raw data is not scaled. Default: False.
adding_hydro_storage_cap_TWh	Additional hydro storage capacity in TWh. If "False", nothing is added. Default: False.
heat_flexibility_Kelvin	Thermal flexibility range in K. The temperature in flexibly heating households can deviate this much from 22°C. Default: 0.5
PVRF_CH_coeficient	Scaling factor for the already available PV capacity in Switzerland prior to any investments. If 1, the PV capacity in 2035 is 30.1 GW and in 2050, it is 35.5 GW. Default: 1.

Setting	Description
battery_cost_factor	Multiplier applied to battery investment costs. If 1 and if the costs have not been modified in the <i>input\cost_operation_invest_data.py</i> file, the investment costs in power capacity will be 44'696 CHF/MW and 38'312 CHF/MW in 2035 and 2050 respectively. The investments costs in storage capacity will be 51'581 CHF/MWh and 44'212 CHF/MWh. Default: 1.

1.2 Secondary settings

Those settings can be added manually to the settings file to change them.

Parameter	Description / Default Value
solver_name	Optimization solver used. Default: gurobi.
allow_investment	Enables investments. Default: True.
merge_some_countries	Merges smaller countries for simplification. Default: True.
minimum.RES.target.CH	Enforces minimal renewable share in CH. Default: False.
RES.CH.coefficient	Scaling factor for renewables in CH before any investments. The unscaled data is saved in <i>input\res_capacities_CH.csv</i> . It has been derived from the original policy dependent data from the Energy Perspectives 2050+ saved in <i>input\res_capacities_CH_old.csv</i> . Default: 1.
PVRF.CH.coefficient	Scaling factor for PV in CH before any investments. The unscaled data is saved in <i>input\res_capacities_CH.csv</i> . It has been derived from the original policy dependent data from the Energy Perspectives 2050+ saved in <i>input\res_capacities_CH_old.csv</i> . Default: 1. Default: 1.
Widon.CH.coefficient	Scaling factor for onshore wind in CH before any investments. The unscaled data is saved in <i>input\res_capacities_CH.csv</i> . It has been derived from the original policy dependent data from the Energy Perspectives 2050+ saved in <i>input\res_capacities_CH_old.csv</i> . Default: 1. Default: 1.

Parameter	Description / Default Value
<code>share_of_available_charging_capacity_for_V2G</code>	The share of their charging capacity that EV owners make available for V2G. They are expected not to provide 100% to conserve their batteries. Default: 0.7.
<code>storage_to_charge_ratio_days</code>	To avoid unrealistic behavior, PTES cannot charge or discharge entirely in less than the given amount of days. If set to 0, the constraint is effectively disabled. Default: 7.
<code>lost_load_cost_mode</code>	Cost modeling of lost load. It can be "single_step" or "multi_step_v1". In case of the latter one, the lost load steps can be defined in <i>input\LostLoadCost\lostLoadCost_multi_step_v1.csv</i> . Default: single_step.
<code>Node_list_setting</code>	List of modeled nodes. Default: ["CH00", "IT00", "DE00", "FR00", "AT00", "DKE1", "DKW1", "BE00", "CZ00", "ES00", "UK00", "HU00", "LU00", "NL00", "PL00", "PT00", "SI00", "SK00", "HR00", "SE01", "SE02", "SE03", "SE04", "NON1", "NOM1", "NOS0",]
<code>NodeDH_list</code>	Set of district heating nodes in the model. It can be set to zero when entering an empty list, i.e. [] For the default, please refer to the <i>scenarios\settings_default.py</i> file.
<code>eu_policy</code>	EU policy setting. "DE" (Distributed Energy) and "GA" (Global Ambition) are available. Default: "GA".
<code>ch_policy</code>	Swiss energy policy configuration from the Energy Perspectives 2050+; ZBA = Zero BAis. Most of the data is also available for "WWB", "ZVA" and "ZVB" but currently, only "ZBA" is available. Default: "ZBA".
<code>winter_limit</code>	A certain net import limit that cannot be exceeded in a given timeframe. If the mode is False, this constraint is not active. Default: {"mode": False, "window": [6553, 2184], "energy_MWh": 5 * 1000000}.
<code>limited_fuels_import_CH_list</code>	List with the fuels with a restricted import, if enabled. Default: ["resmethane", "fossilmethane"].
<code>RES_EU_coefficient</code>	Scaling factor for the non-CH renewable capacities. Default: 1.1.

2 Model Output Files

The model produces several CSV output files and one log file summarizing generation, storage, investments, and system balances. The following overview lists the key outputs, their content, and units. Please note that all files end with *.csv*.

Each file is marked with either *i* for input or *o* for output to indicate whether it serves as an input to the model—typically a parameter—or as an output from the model—typically a variable.

File Name (Input/Output)	Content	Unit
avail_plant(i)	Hourly max utilization for each plant	
BA_max_heating_capacity(i)	Max heating capacity of the heat pumps	MW
BA_th_con(i)	Building archetype hourly thermal consumption	MWh
BA_th_lim(i)	Thermal limit per building archetype; the building archetype thermal limit represents how much the building can be heated or cooled down stepping away from the comfort temperature allowing for the use of thermal inertia	MWh
COP(i)	Coefficient of performance	
cost_data_inv_discharge_slp(o)	Cost of investing in discharge capacity for each technology; only hydrogen has this because the facilities for generating hydrogen and storing hydrogen are considered separately	CHF/MW
cost_inv_dict(o)	Total investment costs of non-thermal plants	CHF
cost_inv_fuel_storage_dict(o)	Total investment costs of fuel storages	CHF
cost_inv_thermal_dict(o)	Total investment costs of thermal plants	CHF
cost_op_dict(o)	Annual operation costs of the individual plants	CHF
cost_op_thermal_dict(o)	Annual operation costs of the individual thermal plants	CHF
curtailment(o)	Occurred curtailment	MWh
curtailmentTh(o)	Occurred thermal curtailment	MWh _{th}
Data_plant_flex_d_within_window(o)	Flexibility data of electrolyzers	MW
demand(i)	Fixed demand	MW
demandDH(i)	Fixed district heating demand	MW _{th}

File Name (Input/Output)	Content	Unit
<code>dsrThDev(o)</code>	Thermal demand side response; records the deviation of the representative building's thermal energy level from the target level for buildings heated by district heating	MWh _{th}
<code>dsrThDev_max(i)</code>	Max thermal demand side response	MWh _{th}
<code>energy_balance_dual(o)</code>	Dual value of the electric energy balance constraint; electricity price	CHF/MWh
<code>energy_balancethermal_dual(o)</code>	Dual value of the thermal energy balance constraint; heat price	CHF/MWh _{th}
<code>energy_max_limit(i)</code>	Max energy capacity that can be invested for certain plants in CH	MWh
<code>energyTH_max_limit(i)</code>	Max energy capacity of thermal energy storage assets that can be invested in CH	MWh _{th}
<code>Export(o)</code>	Utilization of each transmission line	MW
<code>exported(-)</code>	unused	MWh
<code>fuel_consumption_of_fuel(o)</code>	Consumed fuel	MWh
<code>fuel_consumption_of_plant(o)</code>	Consumption per plant	MWh
<code>fuel_import_capacity_annual(i)</code>	Annual fuel import capacity	MWh
<code>fuel_production_capacity_CH_annual(i)</code>	Annual fuel production capacity	MWh
<code>fuel_storage_capacity_annual(o)</code>	Invested storage capacity of fuel storage	MWh
<code>fuel_storage_investment_annual_limit(i)</code>	Max investment allowed in fuel storage capacity	MWh
<code>gen(o)</code>	Hourly generation per plant	MWh
<code>gen_energy_max(i)</code>	Max storable energy or energy available	MWh
<code>gen_energyTh_max(i)</code>	Max storable thermal energy	MWh _{th}
<code>gen_max(o)</code>	Max generation capacity/ Installed capacity; do not confuse with <i>gen_max_limit</i> which is a parameter whereas <i>gen_max</i> is a variable to the model	MW
<code>gen_max_infeedp(i)</code>	Max generation capacity of pre-existing RES plants	MW
<code>gen_max_limit(i)</code>	Upper limit of <i>gen_max</i>	MW
<code>genTh(o)</code>	Hourly thermal generation per plant	MWh _{th}

File Name (Input/Output)	Content	Unit
<code>genTh_max(o)</code>	Max thermal generation capacity/ Installed capacity; do not confuse with <i>genTh_max_limit</i> which is a parameter whereas <i>genTh_max</i> is a variable	MW_{th}
<code>genTh_max_limit(i)</code>	Upper limit of <i>genTh_max</i>	MW_{th}
<code>imported(-)</code>	unused	
<code>infeed(i)</code>	Hourly infeed from pre-existing RES	MWh
<code>inflow(i)</code>	Natural inflow to storage plants	MWh
<code>investment_emax_slp(i)</code>	Storage investment costs	CHF/MWh
<code>investment_emax_slpTh(i)</code>	Thermal storage investment costs	CHF/MWh _{th}
<code>investment_fuel_storage_slp(i)</code>	Fuel storage capacity investment costs	CHF/MWh
<code>investment_genmax_slp(i)</code>	Generation capacity investment costs	CHF/MW
<code>investment_genmax_slpTh(i)</code>	Thermal generation capacity investment costs	CHF/MW
<code>investment_summary(o)</code>	Automatically generated investment overview	
<code>line_trade_price (o)</code>	Only relevant for a different mode of the model	
<code>lostload(o)</code>	Hourly lost load	MWh
<code>lostload_capacity_per_step(i)</code>	Lost load per lost load step; related to <i>lost_load_cost_mode</i>	MWh
<code>lostload_cost_per_step(i)</code>	Lost load costs per lost load step; related to <i>lost_load_cost_mode</i>	CHF/MWh
<code>Map_[a]_[b] (i)</code>	Internally used mapping from a to b	
<code>operation_qdr(i)</code>	Factor in the operation costs' quadratic term of the objective function	CHF/MW ²
<code>operation_slp(i)</code>	Factor in the operation costs' linear term of the objective function	CHF/MW
<code>operation_slpTh(i)</code>	Factor of the thermal operation costs of the objective function	CHF/MW _{th}
<code>outflow(i)</code>	Naturally hourly lost electric energy and consumption of EVs	MWh
<code>P_allinv(i)</code>	All plants/planttypes that the model can invest in	
<code>pmp_max(i)</code>	Charging limit of storages	MW
<code>pumpTh_max(i)</code>	Charging limit of thermal storages	MW _{th}
<code>settings(i)</code>	Settings that were used during this model's execution	

File Name (Input/Output)	Content	Unit
soc(o)	Storages' State of charge	MWh
socTh(o)	Thermal storages' State of charge	MWh _{th}
spill_water(o)	Hourly curtailed hydro inflow	MWh
statistics(o)	Solver termination information	
storage_charge(o)	Hourly charging of storages	MWh
storage_charge_eff_in (i)	Efficiency when charging storages	
storage_charge_eff_out (i)	Efficiency when discharging storages	
storage_chargeTh(o)	Hourly charging of thermal storages	MWh _{th}
storage_soc_dual(o)	Dual variable of the storage balance constraint; water value / storage value	CHF/MWh
storage_start_cond(i)	Relative level at which storages start and end	
sum_generation_plant _energy_limited(i)	Annual upper generation limit of certain technologies	MWh
TES_decayrate(i)	Daily discharge rate of thermal storages	
th_sl(o)	Thermal storage level of households heating flexibly with heat pumps with 0 corresponding to the comfortable reference temperature (22°C)	MWh
trade_cost_dict (o)	Only relevant for a different mode of the model	
V2G_charging_power_rate (i)	Max charging and discharging V2G power rate	MW
V2G_storage_capacity(i)	Max storage capacity of all V2G	MWh
weight_in_objective_fcn (i)	Weight of the individual subscenarios on the objective function	